Credit Card Customer Behavior Analytics

End-to-End Data Analysis & Interactive Power BI Dashboard

Project Report

1. Executive Summary

This project analyzes synthetic credit card customer behavior data using Microsoft Power BI. The objective is to transform customer, spending, payment, credit, rewards, and engagement data into an interactive dashboard that supports customer understanding and business-oriented insights.

The project follows an end-to-end workflow: dataset selection, data understanding, data quality checks, analytical modeling, KPI development, visualization, and insight generation. The final dashboard is organized into Overview, Customer Spending Analysis, Credit & Payment Analysis, and Customer Insights, with an additional experimental page for calculated KPIs using DAX.

2. Project Objective

- Measure the overall customer portfolio and key financial indicators.
- Analyze monthly spending and spending categories.
- Understand payment behavior, credit utilization, outstanding balances, and credit scores.
- Compare customer behavior across card types, age groups, and occupations.
- Explore relationships between credit score, payment ratio, and credit utilization.
- Create calculated customer segments using DAX business rules.
- Present findings through an interactive Power BI dashboard.

3. Dataset Source

The dataset was obtained from Kaggle.

Item	Details
Dataset	Credit Card Customer Behavior Segmentation Dataset
Platform	Kaggle
Source	https://www.kaggle.com/datasets/dharmendrapandit12/credit-card-customer-behavior-segmentation-dataset
Dataset type	Synthetic credit card customer behavior data
Rows	50,000
Columns	30

The dataset description provided for this project states that the data is synthetic and generated using rule-based relationships. Therefore, the findings describe patterns within this dataset and should not be interpreted as conclusions about real bank customers.

4. Data Understanding

The dataset contains customer demographics, income, card characteristics, spending behavior, payment behavior, credit indicators, rewards activity, and mobile-app engagement.

Category	Representative Fields	Purpose
Customer & Demographics	Customer_ID, Age, Gender, Occupation	Describe customer profiles.
Income & Card	Annual_Income, Card_Type, Credit_Limit, Card_Age_Months	Understand financial capacity and card characteristics.
Spending	Monthly_Spending, Monthly_Transactions, Avg_Transaction_Value	Analyze spending behavior.
Spending Categories	Online Shopping, Grocery, Fuel, Dining, Travel, Entertainment, Utility Bills	Identify where customers spend.
Payment & Balance	Outstanding_Balance, Statement_Balance, Payment_Amount, Payment_Ratio	Assess payment behavior and balances.
Credit	Credit_Utilization, Cash_Advance_Amount, EMI_Count, Credit_Score	Assess credit usage and creditworthiness indicators.
Rewards & Engagement	Reward_Points_Earned, Reward_Points_Redeemed, Mobile_App_Login	Measure rewards and digital engagement.
International Activity	International_Transactions	Measure international card usage.

5. Data Quality & Cleaning

The dataset was checked for basic data-quality issues before analysis. The checks performed showed 50,000 rows and 30 columns, with no empty cells and no duplicate rows.

Check	Result
Missing / empty cells	None identified
Duplicate rows	None identified
Basic data consistency	Reviewed before analysis
Dataset size	50,000 rows × 30 columns

Because the dataset was already clean based on these checks, no mass deletion or missing-value imputation was required. The analysis then focused on measures, grouping fields, and calculated KPIs in Power BI.

6. Dashboard & Analytical Approach

6.1 Overview

Business question: What does the overall customer portfolio look like?

- Total Customers
- Total Monthly Spending
- Total Outstanding Balance
- Total Reward Points Earned
- Average Annual Income
- Average Payment Ratio
- Average Credit Score
- Monthly Spending by Card Type
- Customer Distribution by Card Type

- Average Credit Score by Card Type
- Payment Ratio vs. Credit Utilization

6.2 Customer Spending Analysis

Business question: How do customers spend?

- Average Monthly Spending
- Average Transaction Value
- Average Monthly Transactions
- Total spending by major spending category
- Average Monthly Spending by Age Group
- Average Monthly Spending by Card Type
- Top 3 Occupations by Total Spending
- Spending Category Distribution

6.3 Credit & Payment Analysis

Business question: How do customers manage credit and payments?

- Average Credit Limit
- Average Credit Utilization
- Average Outstanding Balance
- Average Statement Balance
- Average Payment Amount
- Average Payment Ratio
- Average Cash Advance
- Average EMI Count
- Average Credit Score
- Credit Score vs. Credit Utilization
- Payment Ratio vs. Credit Score
- Credit Score Distribution
- Credit Utilization by Card Type

6.4 Customer Insights

Business question: What customer characteristics are associated with behavior and engagement?

- Average Mobile App Login
- Average Reward Points Earned
- Average Reward Points Redeemed
- Average International Transactions
- Average Card Age
- Age vs. Spending
- Income vs. Credit Limit
- Reward Points by Card Type
- App Usage (Logins) by Age Group
- Customers by Occupation
- Customers by Card Type

7. Calculated KPIs & Customer Segmentation

7.1 High Value Customers

A customer is classified as High Value when all three conditions are satisfied:

- Annual Income is above the overall customer average.
- Monthly Spending is above the overall customer average.
- Payment Ratio is at least 90%.

This segment combines financial capacity, spending activity, and strong payment behavior.

7.2 Active Customers

A customer is classified as Active when both conditions are satisfied:

- Monthly Transactions are above the average number of monthly transactions.
- Mobile App Login is above the average number of logins.

This segment combines transactional activity with digital engagement.

7.3 At-Risk Customers

The At-Risk segment is based on a combination of indicators:

- High Credit Utilization.
- Low Payment Ratio.
- High Outstanding Balance.
- Low Credit Score.

The purpose is to identify customers showing multiple signals that may indicate higher financial risk within the synthetic dataset.

8. Key Findings

- Average monthly spending showed only modest change across age groups, suggesting that age alone does not strongly differentiate average spending in this dataset.
- Card types showed noticeable differences in average monthly spending, making card type useful for comparing customer spending behavior.
- Business Owners, Private Employees, and Self-Employed customers appeared among the top occupations by total spending.
- Grocery represented the largest share of the spending-category distribution, followed by Utility Bills and Online Shopping.
- The Credit Score vs. Credit Utilization scatter plot showed a downward trend line, indicating a negative relationship between the two variables in the dataset.
- The Payment Ratio vs. Credit Score scatter plot showed an upward trend line, indicating a positive relationship between the two variables in the dataset.
- Credit score values were concentrated mainly within the middle-to-higher ranges shown in the distribution chart.
- Average credit utilization across card types appeared relatively close, with differences between card types being limited compared with differences observed in spending.

Interpretation note: trend lines show association within the synthetic dataset and do not establish causation.

9. Recommendations

- Use customer spending and card-type patterns to support targeted product and rewards strategies.
- Consider tailored engagement campaigns for customers with high spending and strong payment behavior.
- Monitor customers showing a combination of high credit utilization, low payment ratio, high outstanding balance, and low credit score.
- Use mobile-app engagement to identify opportunities for personalized in-app communication.
- Analyze reward earning and redemption patterns by card type to evaluate rewards-program alignment.
- Use age, occupation, income, and card type together rather than relying on a single demographic variable.

10. Power BI Dashboard Design

The final dashboard uses KPI cards for high-level metrics and bar, column, line/area, donut, and scatter visuals according to the analytical question. Consistent formatting, colors, titles, and layouts were used across pages to create a unified user experience.

The dashboard moves from portfolio-level performance to spending behavior, credit and payment behavior, and finally customer insights and segmentation.

11. Tools Used

- Microsoft Excel — initial data inspection and preparation.
- Microsoft Power BI — data modeling, DAX calculations, interactive visualizations, and dashboard development.
- DAX — calculated measures and customer segmentation logic.
- Kaggle — dataset source.

12. Conclusion

This project demonstrates an end-to-end approach to customer behavior analytics using Power BI. Starting from a 50,000-row synthetic credit card dataset, the project covered data understanding, quality checks, KPI development, business-oriented analysis, interactive visualization, relationship analysis, and DAX-based customer segmentation.

The final dashboard turns customer-level financial and behavioral fields into a structured analytical story covering spending behavior, credit and payment patterns, customer engagement, and potential customer segments.